CLASS 12 · JEE MAIN & ADVANCED · PHYSICS

Electric Charges and Coulomb's Law

Detailed notes, step by step, with worked examples

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Written to go with the class “Electrostatic Charges & Coulomb's Force” (Vedantu JEE English, YouTube).
These are original notes on the same topics, not a transcript of the class.
Every worked answer is checked by code.

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How to use these notes

- Read a section, then cover the worked example and try it yourself.
- Grey boxes in examples are the final answers. Coloured boxes: **orange** = key idea, **blue** = formula, **red** = common mistake, **green** = tip.
- Finish with the practice questions. Answers are on the last page.

1 What is electric charge?

Rub a plastic comb on dry hair and it starts picking up bits of paper. Something has changed on the comb. We say it has become **charged**. Electric charge is a basic property of matter, just like mass. Some particles have it, some don't, and it is what makes electric forces happen.

There are exactly **two kinds** of charge. Benjamin Franklin named them **positive (+)** and **negative (−)**. The names are just labels. What matters is how they behave:



Like charges push apart

Unlike charges pull together

The one rule that never changes: like charges repel, unlike charges attract.

Remember

- Like charges **repel**. Unlike charges **attract**.
- The SI unit of charge is the **coulomb (C)**.
- In everyday problems, charges are usually tiny: $1\ \mu\text{C} = 10^{-6}$ C and $1\ \text{nC} = 10^{-9}$ C.

Where does charge come from?

Every atom has a small heavy nucleus (protons and neutrons) with electrons around it. Protons carry positive charge, electrons carry an equal amount of negative charge, and neutrons carry none.

Particle	Charge	Mass
Electron	$-e = -1.6 \times 10^{-19}$ C	9.1×10^{-31} kg
Proton	$+e = +1.6 \times 10^{-19}$ C	1.67×10^{-27} kg
Neutron	0	1.67×10^{-27} kg

A neutral atom has as many electrons as protons, so the charges cancel. A body becomes **negatively charged** when it gains extra electrons, and **positively charged** when it loses some. Only electrons move around in everyday charging. The protons stay locked in the nucleus.

Common slip

A positively charged body has **lost electrons**. It has not gained protons.

2 Properties of electric charge

1. Charge is additive (it's a scalar)

The total charge of a system is just the algebraic sum of all the charges in it, with their signs. For example, charges $+3$ C, -5 C and $+1$ C add up to -1 C. Charge has no direction, so it is a **scalar**.

2. Charge is quantised

Charge always comes in whole-number multiples of the smallest charge, e. You can't have half an electron's charge on a body.

Quantisation of charge

$$q = \pm ne, \qquad n = 1, 2, 3, \ldots, \qquad e = 1.6 \times 10^{-19}\ \mathrm{C}$$

Example 1 · Counting electrons

A body has a charge of $-3.2\ \mu\mathrm{C}$. How many extra electrons does it carry?

SOLUTION, STEP BY STEP

1. Use $q = ne$, so $n = \dfrac{q}{e}$ (we take the size of q; the minus sign just tells us the extra particles are electrons).
2. $n = \dfrac{3.2 \times 10^{-6}}{1.6 \times 10^{-19}}$.
3. Divide the numbers and the powers separately: $\dfrac{3.2}{1.6} = 2$ and $\dfrac{10^{-6}}{10^{-19}} = 10^{13}$.

Answer: $n = 2 \times 10^{13}$ extra electrons.

Because e is so small, quantisation is invisible in everyday life. A charge of just $1\ \mu\mathrm{C}$ is already about 6×10^{12} electrons, so charge looks continuous.

3. Charge is conserved

Charge can't be created or destroyed. It can only move from one body to another. When you rub glass with silk, the glass loses exactly as many electrons as the silk gains. The total charge of an isolated system never changes.

4. Charge is invariant

The charge on a particle is the same no matter how fast it moves. (Mass, by contrast, changes at speeds close to light.)

Charge vs mass, side by side

Charge	Mass
Can be positive, negative or zero	Always positive
Quantised: always a multiple of e	Not quantised in this sense
Same at any speed	Changes at very high speeds
Like charges repel, unlike attract	Masses always attract

3 Conductors, insulators and earthing

- **Conductors** let charge move through them easily, because some electrons are free to wander. Metals, the human body and the Earth are conductors.
- **Insulators** hold charge where it is put. Their electrons are tightly bound. Glass, plastic, rubber, dry wood and silk are insulators.
- **Earthing (grounding)** means connecting a conductor to the Earth. The Earth is so big that it can give or take any amount of charge, so an earthed conductor shares away its extra charge.

Why it matters

Charge given to a conductor spreads out over its outer surface. Charge given to an insulator stays at the spot where it was placed.

4 Three ways to charge a body

1. By friction (rubbing)

Rubbing two different materials moves electrons from one to the other. Both end up with **equal and opposite** charges.

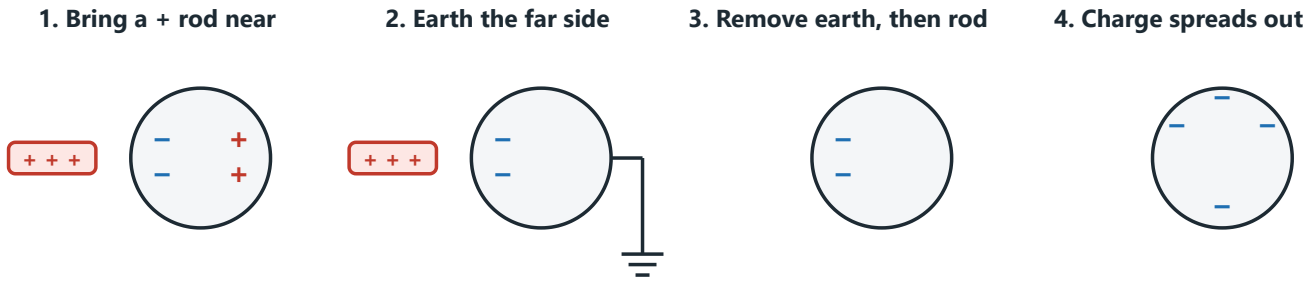
Rub this…	…with this	Result
Glass rod	Silk	Glass $+$, silk $-$
Ebonite (or plastic) rod	Fur or wool	Ebonite $-$, fur $+$

2. By conduction (touching)

Touch a charged body to a neutral conductor and some charge flows across. The neutral body ends up with the **same sign** of charge, and the first body loses some of its charge.

3. By induction (without touching)

This is the clever one. You charge a conductor without ever touching it with the charged object:



Charging a sphere negatively by induction with a positive rod.

1. Bring a positive rod near a neutral metal sphere. The sphere's free electrons are pulled towards the rod, so the near side becomes negative and the far side positive.
2. Earth the sphere. Electrons flow up from the Earth and cancel the positive charge on the far side.
3. Remove the earth connection **first**, then take the rod away.
4. The extra electrons spread over the whole sphere. It is now negatively charged.

> **Induction in one line**
>
> Induction always gives the body the **opposite** sign to the charge that induced it, and the inducing body loses nothing.

> **Order matters**
>
> If you take the rod away **before** removing the earth, the electrons simply flow back to the Earth and the sphere ends up neutral.

5 Coulomb's law

In 1785 Charles-Augustin de Coulomb measured the force between charged bodies with a torsion balance. He found a simple rule:

> **Coulomb's law in words**
>
> The force between two point charges is **directly proportional to the product of the charges** and **inversely proportional to the square of the distance** between them. It acts along the line joining them.



Two point charges a distance r apart. Each pushes the other with the same size of force.

Coulomb's law

$$F = k\,\frac{q_1 q_2}{r^2}, \qquad k = \frac{1}{4\pi\varepsilon_0} = 9\times 10^9\ \mathrm{N\,m^2/C^2}$$

$\varepsilon_0 = 8.85\times 10^{-12}\ \mathrm{C^2/(N\,m^2)}$ is the **permittivity of free space**. (More precisely $k = 8.99\times 10^9$; JEE uses 9×10^9.)

Reading the formula

- **Product of charges.** Double one charge and the force doubles. Double both and it becomes four times as big.
- **Inverse square.** Double the distance and the force drops to **one quarter**, not one half. Halve the distance and it becomes four times as big.
- **Sign tells the type.** Put the charges in with their signs: a positive answer means repulsion, a negative answer means attraction. (For the size of the force, use the magnitudes.)

When does it apply?

- To **point charges**, meaning bodies much smaller than the distance between them.
- To charges **at rest** (that's why this topic is called electro**statics**).

Vector form

Forces have direction, so we can write the law as a vector. Let $\vec{r}_{12}$ be the vector from q_1 to q_2, with unit vector $\hat{r}_{12}$. The force on q_2 due to q_1 is

Vector form

$$\vec{F}_{21} = k\,\frac{q_1 q_2}{r^2}\,\hat{r}_{12} = k\,\frac{q_1 q_2}{r^3}\,\vec{r}_{12}$$

If $q_1 q_2 > 0$ the force on q_2 points away from q_1 (repulsion); if $q_1 q_2 < 0$ it points towards q_1 (attraction). And $\vec{F}_{12} = -\vec{F}_{21}$.

Example 2 · Force between two charges

Find the force between $+3\ \mu\text{C}$ and $-2\ \mu\text{C}$ placed 30 cm apart in vacuum.

SOLUTION, STEP BY STEP

1. Convert to SI: $q_1 = 3 \times 10^{-6}$ C, $q_2 = 2 \times 10^{-6}$ C (size), $r = 0.3$ m.
2. $F = k\,\dfrac{q_1 q_2}{r^2} = \dfrac{(9 \times 10^9)(3 \times 10^{-6})(2 \times 10^{-6})}{(0.3)^2}$.
3. Top: $9 \times 3 \times 2 = 54$ and $10^9 \times 10^{-6} \times 10^{-6} = 10^{-3}$, so 54×10^{-3}. Bottom: 0.09.
4. $F = \dfrac{54 \times 10^{-3}}{0.09} = 0.6$ N.
5. The charges have opposite signs, so the force is attractive.

Answer: $F = 0.600$ N, attractive.

6 Coulomb's law in a medium

Put the same two charges inside oil or water instead of vacuum, and the force between them becomes weaker. The material partly shields the charges from each other. How much weaker is measured by its **dielectric constant** K (also written ε_r, the relative permittivity).

Force in a medium

$$F_{\text{medium}} = \frac{1}{4\pi\varepsilon_0 K}\,\frac{q_1 q_2}{r^2} = \frac{F_{\text{vacuum}}}{K}, \qquad \varepsilon = K\varepsilon_0$$

Medium	Dielectric constant K
Vacuum	1 (by definition)
Air	≈ 1 (1.0006)
Water	≈ 80
Metal (conductor)	∞ (no field inside)

Example 3 · Same charges, now in a liquid

The two charges of the last example are placed, at the same distance, in a liquid with $K = 4$. Find the new force.

SOLUTION, STEP BY STEP

1. In a medium the force is divided by K: $F' = \dfrac{F}{K}$.
2. $F' = \dfrac{0.6}{4}$.

Answer: $F' = 0.150$ N, still attractive.

7 Electric force vs gravitational force

Both forces follow an inverse-square law, but they are wildly different in strength.

Electric force	Gravitational force
$F = k\dfrac{q_1 q_2}{r^2}$	$F = G\dfrac{m_1 m_2}{r^2}$
Can attract or repel	Only attracts
Depends on the medium	Does not depend on the medium
Very strong	Very weak

Example 4 · How much stronger?

Compare the electric and gravitational forces between an electron and a proton. Take $G = 6.67 \times 10^{-11}$, $m_e = 9.1 \times 10^{-31}$ kg, $m_p = 1.67 \times 10^{-27}$ kg.

SOLUTION, STEP BY STEP

1. Both forces have r^2 in the bottom, so in the ratio the distance cancels: $\dfrac{F_e}{F_g} = \dfrac{ke^2}{Gm_e m_p}$.
2. Top: $(9 \times 10^9)(1.6 \times 10^{-19})^2 = 2.304 \times 10^{-28}$.
3. Bottom: $(6.67 \times 10^{-11})(9.1 \times 10^{-31})(1.67 \times 10^{-27}) \approx 1.014 \times 10^{-67}$.
4. Divide: $\dfrac{2.304 \times 10^{-28}}{1.014 \times 10^{-67}}$.

Answer: $\dfrac{F_e}{F_g} \approx 2.27 \times 10^{39}$. The electric force wins by about 10^{39}!

So why does gravity rule the sky?

Big bodies are almost exactly neutral, so their electric forces cancel out. Gravity never cancels, because mass is always positive.

8 Newton's third law and the superposition principle

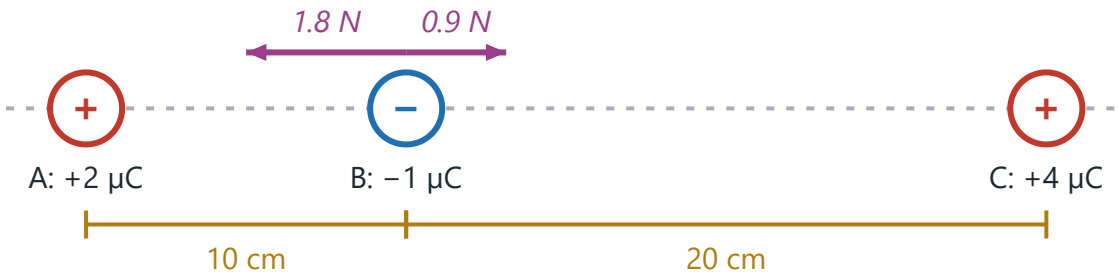
The force q_1 puts on q_2 is always equal and opposite to the force q_2 puts on q_1, **even if the charges are different sizes**. Both use the same product $q_1 q_2$ in Coulomb's law.

Superposition principle

When many charges act on one charge, each pair interacts **as if the others were not there**. The net force is the **vector sum** of the separate forces:

$$\vec{F}_{\text{net}} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \dots$$

How to use it, every time: (1) find the size of each force with Coulomb's law; (2) draw its direction (away for like charges, towards for unlike); (3) add the arrows as vectors.



Forces on B from A and from C.

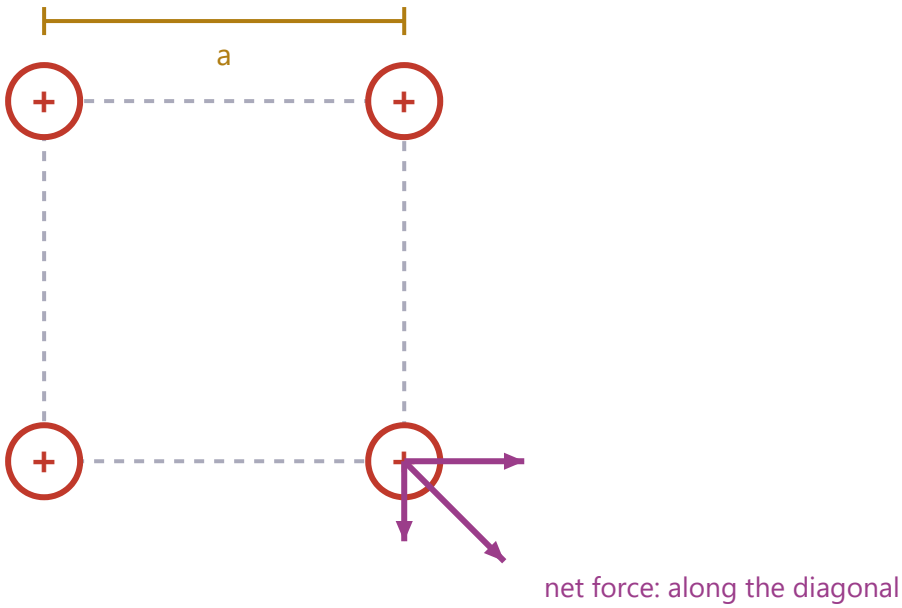
Example 5 · Charges on a line

$A = +2\ \mu\text{C}$ at $x = 0$, $B = -1\ \mu\text{C}$ at $x = 10$ cm, $C = +4\ \mu\text{C}$ at $x = 30$ cm. Find the net force on B.

SOLUTION, STEP BY STEP

1. Force from A: $\dfrac{(9\times10^{9})(2\times10^{-6})(1\times10^{-6})}{(0.1)^2} = 1.80$ N. A and B are unlike, so it pulls B **towards A** (left).
2. Force from C: $\dfrac{(9\times10^{9})(4\times10^{-6})(1\times10^{-6})}{(0.2)^2} = 0.900$ N. Unlike again, so it pulls B **towards C** (right).
3. They act along the same line in opposite directions, so subtract: $1.8 - 0.9$.

Answer: 0.900 N, towards A.



Four equal charges on a square: forces on the bottom-right corner.

Example 6 · Four charges on a square

Equal charges $q = 1\ \mu\text{C}$ sit on the corners of a square of side $a = 10$ cm. Find the net force on any one charge.

SOLUTION, STEP BY STEP

1. The two neighbours are a distance a away. Each pushes with $F_1 = \dfrac{kq^2}{a^2} = 0.900$ N, at right angles to each other.
2. Two equal forces at 90° add to $\sqrt{2}\,F_1$, pointing along the diagonal.
3. The opposite corner is $\sqrt{2}\,a$ away. Its push is $\dfrac{kq^2}{(\sqrt{2}a)^2} = \dfrac{F_1}{2}$, also along the diagonal.
4. Everything now lies along the diagonal, so just add: $F = \sqrt{2}\,F_1 + \tfrac{1}{2}F_1 = \left(\sqrt{2} + \tfrac{1}{2}\right)\dfrac{kq^2}{a^2}$.

Answer: $F = \left(\sqrt{2} + \tfrac{1}{2}\right)(0.900) \approx 1.72$ N, along the diagonal, away from the centre.

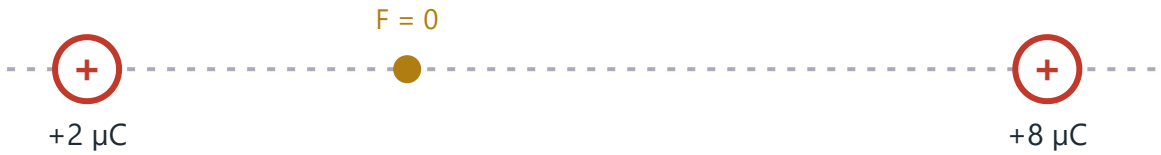
Handy result

Three equal charges q on an equilateral triangle of side a: net force on each is $\sqrt{3}\,\dfrac{kq^2}{a^2}$, pointing away from the centre.

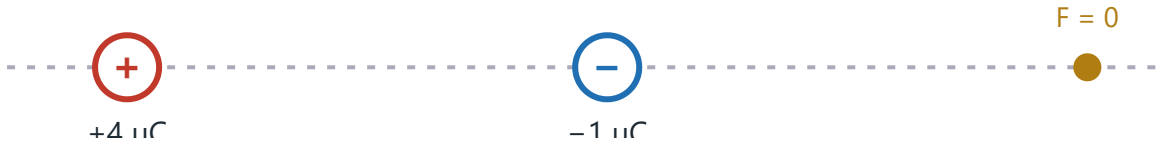
9 Where is the net force zero? (Equilibrium)

A very common JEE question: two fixed charges, and you must find where a third charge would feel **no net force**. This point is called the **null point**.

Like charges: null point between them, closer to the smaller one



Unlike charges: null point outside, beyond the smaller one



Where the null point lies, for like and unlike charges.

Where to look

- **Like charges** (both + or both −): the null point is **between** them, closer to the **smaller** charge.
- **Unlike charges**: the null point is **outside**, on the side of the **smaller** charge (beyond it).
- The size and sign of the test charge **don't matter**. It cancels out of the equation.

Example 7 · Like charges

$+2\ \mu\text{C}$ and $+8\ \mu\text{C}$ are 30 cm apart. Where between them is the net force on a test charge zero?

SOLUTION, STEP BY STEP

1. Let the point be x cm from the $2\ \mu\text{C}$ charge, so $(30 - x)$ cm from the $8\ \mu\text{C}$ charge.
2. For zero net force the two pushes must be equal: $\dfrac{k\,(2)\,q}{x^2} = \dfrac{k\,(8)\,q}{(30-x)^2}$. Cancel k and q.
3. Take square roots: $\dfrac{\sqrt{2}}{x} = \dfrac{\sqrt{8}}{30-x}$, and $\sqrt{8} = 2\sqrt{2}$, so $30 - x = 2x$.
4. $3x = 30$, so $x = 10$ cm.

Answer: 10.0 cm from the $+2\ \mu\text{C}$ charge.

Example 8 · Unlike charges

$+4\ \mu\text{C}$ and $-1\ \mu\text{C}$ are 30 cm apart. Where is the null point?

SOLUTION, STEP BY STEP

1. Between them both forces on a test charge point the same way, so they can't cancel there. It must be outside, beyond the **smaller** charge ($-1\ \mu\text{C}$).
2. Let it be x cm beyond the $-1\ \mu\text{C}$ charge, so $(30 + x)$ cm from the $+4\ \mu\text{C}$ charge.
3. $\dfrac{4}{(30+x)^2} = \dfrac{1}{x^2}$. Square roots: $\dfrac{2}{30+x} = \dfrac{1}{x}$, so $2x = 30 + x$.

Answer: $x = 30.0$ cm beyond the $-1\ \mu\text{C}$ charge (that is, 60 cm from the $+4\ \mu\text{C}$ charge).

Putting all three charges in equilibrium

Example 9 · A third charge that balances everything

Charges q and $4q$ are a distance d apart. What charge Q, placed where, makes **all three** charges stay in equilibrium?

SOLUTION, STEP BY STEP

1. For Q itself: like the example above, Q must sit between them at $\dfrac{d}{3}$ from q (here $\sqrt{q} : \sqrt{4q} = 1 : 2$).
2. For q: the push from $4q$ (repulsion, size $\dfrac{k\,4q^2}{d^2}$) must be cancelled by a pull from Q, so Q must be **negative**.
3. Balance on q: $\dfrac{k\,q\,|Q|}{(d/3)^2} = \dfrac{k\,q\,(4q)}{d^2}$, so $9|Q| = 4q$, giving $|Q| = \dfrac{4q}{9}$.
4. Check on $4q$: pull from Q is $\dfrac{k(4q)(4q/9)}{(2d/3)^2} = \dfrac{4kq^2}{d^2}$, which equals the push from q. Balanced.

Answer: $Q = -\dfrac{4q}{9}$, placed at $\dfrac{d}{3}$ from q.

Stable or unstable?

A positive test charge at the null point between two positive charges is **stable** if you nudge it along the line (it gets pushed back), but **unstable** if you nudge it sideways.

10 Touching identical spheres (sharing charge)

When two **identical** conducting spheres touch, the total charge is shared **equally** between them. Then they are moved apart again.

Sharing between identical spheres

$$q_{\text{each}} = \frac{q_1 + q_2}{2} \quad \text{(add with signs!)}$$

Example 10 · Force before and after touching

Two identical spheres have $+8\ \mu\text{C}$ and $-2\ \mu\text{C}$. They touch and are put back at the same distance. Compare the forces.

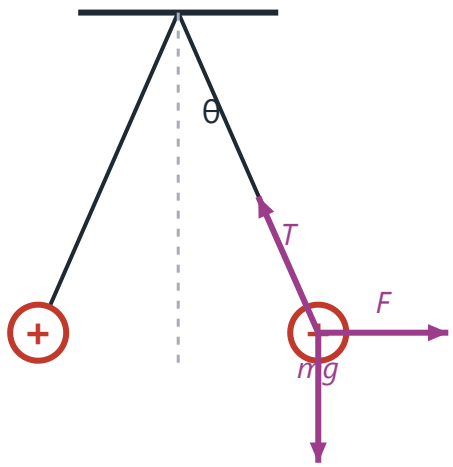
SOLUTION, STEP BY STEP

1. After touching, each has $\dfrac{8 + (-2)}{2} = +3\ \mu\text{C}$.
2. Before: $F \propto 8 \times 2 = 16$ (attractive, opposite signs).
3. After: $F' \propto 3 \times 3 = 9$ (repulsive, same signs).

Answer: $\dfrac{F'}{F} = \dfrac{9}{16}$, and the force changes from **attraction** to **repulsion**.

11 Hanging charged balls

Two small identical balls, each with charge q and mass m, hang from the same point on threads. They repel and settle with each thread at an angle θ to the vertical.



Forces on one ball: tension T along the thread, weight mg down, electric force F sideways.

1. Three forces act on each ball: tension T (along the thread), weight mg (down), and the electric push F (horizontal, away from the other ball).
2. The ball is at rest, so the forces balance. Vertically: $T\cos\theta = mg$.
3. Horizontally: $T\sin\theta = F$.
4. Divide the two equations: T cancels.

Equilibrium of a hanging charged ball

$$\tan\theta = \frac{F}{mg} = \frac{kq^2}{r^2\,mg}$$

Example 11 · Finding the angle

Each ball has mass 0.9 g and charge $0.1\ \mu\text{C}$. They settle 10 cm apart. Find θ ($g = 10$ m/s²).

SOLUTION, STEP BY STEP

1. $F = \dfrac{(9\times10^{9})(10^{-7})^2}{(0.1)^2} = 9.00\times10^{-3}$ N.
2. $mg = (0.9\times10^{-3})(10) = 9.00\times10^{-3}$ N.
3. $\tan\theta = \dfrac{F}{mg} = 1.00$.

Answer: $\theta = 45.0^\circ$.

Dipping them in a liquid

In a liquid of density σ and dielectric constant K, two things change: the electric force drops to F/K, and buoyancy reduces the weight to $mg\,(1-\sigma/\rho)$, where ρ is the density of the balls. The angle stays the same if both shrink by the same factor:

Same angle in a liquid

$$K = \frac{\rho}{\rho - \sigma}$$

For example, balls of density 1600 kg/m³ in a liquid of density 800 kg/m³ keep the same angle when $K = 2.00$.

12 When is the force largest?

Example 12 · Splitting a charge

A charge Q is split into two parts, q and $Q - q$, placed a fixed distance apart. For what q is the repulsion largest?

SOLUTION, STEP BY STEP

1. $F = k\,\dfrac{q(Q-q)}{r^2}$. Only the product $q(Q-q)$ changes.
2. For the largest value, set the derivative to zero: $\dfrac{d}{dq}\big[q(Q-q)\big] = Q - 2q = 0$.
3. So $q = \dfrac{Q}{2}$. (The second derivative is $-2 < 0$, so it's a maximum.)

Answer: $q = \dfrac{Q}{2}$: split the charge equally. (A numerical check gives $0.500\,Q$.)

13 Common mistakes to avoid

- **Forgetting to convert units.** μC to C ($\times 10^{-6}$) and cm to m ($\times 10^{-2}$) before using $k = 9 \times 10^9$.
- **Forgetting to square r.** The force goes as $1/r^2$, not $1/r$.
- **Adding forces as plain numbers.** Forces are vectors. Add them as arrows, unless they lie on one line.
- **Thinking a bigger charge feels a bigger force.** In a pair, both charges feel the same size of force.
- **Searching for the null point on the wrong side.** Like charges: between them. Unlike charges: outside, beyond the smaller one.
- **Sharing charge without signs.** $+8$ and -2 share to $+3$ each, not $+5$.
- **Using the medium wrongly.** In a medium the force is F/K, smaller than in vacuum.

14 Formula sheet

Idea	Formula
Quantisation	$q = ne,\ e = 1.6 \times 10^{-19}$ C
Coulomb's law	$F = k\dfrac{q_1 q_2}{r^2},\ k = \dfrac{1}{4\pi\varepsilon_0} = 9 \times 10^9$
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ \mathrm{C^2/N\,m^2}$
Vector form	$\vec{F}_{21} = k\dfrac{q_1 q_2}{r^3}\vec{r}_{12}$
In a medium	$F_m = \dfrac{F_0}{K}$
Superposition	$\vec{F}_{\text{net}} = \sum \vec{F}_i$
Null point (like charges)	$x = \dfrac{d\sqrt{q_1}}{\sqrt{q_1}+\sqrt{q_2}}$ from q_1
Identical spheres touching	$q_{\text{each}} = \dfrac{q_1 + q_2}{2}$
Hanging balls	$\tan\theta = \dfrac{F}{mg}$
Same angle in a liquid	$K = \dfrac{\rho}{\rho - \sigma}$
Square of equal charges	$F = \left(\sqrt{2} + \tfrac{1}{2}\right)\dfrac{kq^2}{a^2}$
Equilateral triangle	$F = \sqrt{3}\,\dfrac{kq^2}{a^2}$

15 Practice questions

Try each one before you look at the answers on the next page.

1. Two charges of $+5\ \mu\mathrm{C}$ each are 50 cm apart in air. Find the force between them and say whether it attracts or repels.
2. How many electrons must be removed from a neutral body to give it a charge of $+1.6\ \mu\mathrm{C}$?
3. If the distance between two charges is halved, what happens to the force?
4. Charges $+1\ \mu\mathrm{C}$ and $+9\ \mu\mathrm{C}$ are 40 cm apart. Where between them is the net force on a test charge zero?
5. Identical spheres carry $+6\ \mu\mathrm{C}$ and $-2\ \mu\mathrm{C}$. They touch and go back to the same distance. Find each new charge and the ratio $F_{\text{after}}/F_{\text{before}}$.

6. Three charges of $+2\ \mu\text{C}$ sit on the corners of an equilateral triangle of side 20 cm. Find the net force on any one of them.
7. Two charges are moved from air into water ($K = 80$) at the same distance. What happens to the force?
8. A charge Q is split into two parts, q and $Q - q$. For what q is the force between the parts largest?

16 Answers

1. 0.900 N, repulsive.
2. 10^{13} electrons.
3. It becomes 4 times as large.
4. 10.0 cm from the $1\ \mu\text{C}$ charge.
5. $+2\ \mu\text{C}$ each; ratio $\dfrac{2 \times 2}{6 \times 2} = \dfrac{1}{3}$ (and it changes from attraction to repulsion).
6. $\sqrt{3}\,\dfrac{kq^2}{a^2} = 1.56$ N, pointing away from the centre of the triangle.
7. It becomes $\frac{1}{80}$ of the force in air.
8. $q = \dfrac{Q}{2}$.